

HOWE SOUND PAM ROCKS, BC, Canada

WMO: 712110

Lat: 49.488N

Lon: 123.300W

Elev: 7

StdP: 101.24

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -2.5	(c) -0.7	(d) -11.6	(e) 1.4	(f) -0.5	(g) -8.9	(h) 1.8	(i) 1.6	(j) 21.7	(k) -1.4	(l) 19.4	(m) 2.1	(n) 13.3	(o) 20	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.2	(c) 24.7	(d) 18.8	(e) 23.1	(f) 18.2	(g) 21.9	(h) 17.7	(i) 19.8	(j) 23.1	(k) 18.9	(l) 22.1	(m) 18.2	(n) 21.2	(o) 3.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.4	(b) 13.3	(c) 21.9	(d) 17.6	(e) 12.6	(f) 21.0	(g) 16.8	(h) 12.0	(i) 20.3	(j) 56.7	(k) 23.1	(l) 53.8	(m) 22.1	(n) 51.5	(o) 21.3	(p) 25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.7	13.4	-3.9	27.9	2.1	2.1	-5.5	29.4	-6.7	30.7	-7.9	31.8	-9.4	33.4
			-6.0	21.0	2.3	1.0	-7.6	21.8	-9.0	22.4	-10.2	23.0	-11.9	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	5.2	5.9	7.3	10.1	13.7	16.1	18.7	18.9	15.9	11.2	7.3	4.8		
	DBStd	5.59	2.72	2.45	2.31	2.31	2.64	2.50	2.32	1.99	2.17	2.07	2.62	2.64		
	HDD10.0	637	148	116	87	26	2	0	0	0	0	11	86	161		
	HDD18.3	2652	406	349	341	248	146	75	22	17	77	221	331	419		
	CDD10.0	1110	0	1	4	29	116	184	270	274	178	48	4	1		
	CDD18.3	82	0	0	0	0	1	9	34	33	5	0	0	0		
	CDH23.3	123	0	0	0	0	2	13	63	41	4	0	0	0		
	CDH26.7	13	0	0	0	0	0	1	7	6	0	0	0	0		
(14) Wind	WSAvg	5.0	7.0	6.6	5.8	4.7	4.2	3.8	3.4	3.4	3.6	4.7	6.4	7.3		
(15) Precipitation	PrecAvg	1932	276	181	174	127	94	77	53	62	82	218	290	284		
	PrecMax	2315	455	347	317	252	137	130	99	273	185	373	501	354		
	PrecMin	1605	165	44	69	44	32	30	11	15	25	72	154	192		
	PrecStd	195	68	69	49	53	29	31	23	63	42	75	91	48		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.5	14.5	18.7	22.4	25.3	27.4	27.1	23.2	17.9	14.4	12.6		
		MCWB	10.0	9.6	10.3	12.2	14.9	17.2	19.8	20.0	18.1	13.5	12.2	10.6		
	2%	DB	10.3	10.6	12.5	16.2	20.0	22.7	25.0	24.2	21.1	16.0	12.7	10.1		
		MCWB	8.5	7.9	9.4	11.7	14.8	16.7	19.5	18.9	17.0	13.0	10.7	8.0		
	5%	DB	9.3	9.6	11.3	14.8	18.5	21.0	23.4	22.8	19.9	14.9	11.4	8.7		
		MCWB	7.7	7.3	8.5	11.0	14.3	16.2	18.6	18.3	16.3	12.3	9.7	7.0		
	10%	DB	8.3	8.7	10.2	13.4	17.3	19.7	22.0	21.7	18.8	13.9	10.4	7.8		
		MCWB	6.9	6.8	7.8	10.1	13.6	15.6	18.0	17.9	15.7	11.7	8.8	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.4	11.2	13.6	16.5	18.7	21.6	20.6	19.4	15.3	12.9	10.9		
		MCDB	10.8	11.8	13.2	16.3	19.8	23.0	25.3	24.6	22.0	16.9	13.9	12.9		
	2%	WB	8.8	8.6	10.0	12.5	15.5	17.6	20.2	19.6	18.1	14.0	11.0	8.5		
		MCDB	9.7	9.8	11.9	15.3	18.6	21.3	23.7	22.8	20.4	15.4	12.2	9.7		
	5%	WB	7.9	7.8	9.1	11.5	14.7	16.7	19.2	18.9	17.1	12.9	10.0	7.4		
		MCDB	8.8	9.0	10.8	14.0	17.7	20.2	22.5	21.8	19.2	14.2	11.1	8.4		
	10%	WB	7.1	7.1	8.3	10.6	13.9	15.8	18.3	18.2	16.3	12.1	9.1	6.7		
		MCDB	8.0	8.3	9.9	12.9	16.7	19.1	21.5	21.0	18.4	13.5	10.0	7.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	3.6	4.2	4.9	5.1	5.2	5.7	5.2	4.6	3.6	3.1	2.8		
		MCDBR	4.3	4.5	5.4	6.8	6.9	7.4	7.9	7.0	5.9	4.8	4.6	4.4		
	5% WB	MCWBR	3.0	2.9	3.6	4.3	4.5	4.4	4.3	4.1	4.0	3.2	3.4	3.1		
		MCWBR	4.0	3.9	4.7	5.6	5.8	6.2	6.9	6.1	5.4	4.1	4.5	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.310	0.333	0.366	0.384	0.371	0.365	0.368	0.349	0.339	0.321	0.304		
		taud	2.367	2.443	2.401	2.349	2.357	2.422	2.460	2.468	2.524	2.530	2.476	2.404		
	Ebn at Noon		724	842	878	880	875	888	888	869	853	799	725	691		
		Edh at Noon	67	78	97	113	117	111	105	100	86	72	61	57		
(44) All-Sky Solar Radiation	RadAvg		0.83	1.68	2.47	3.89	5.14	5.47	6.18	5.26	3.60	1.98	0.97	0.66		
	RadStd		0.15	0.27	0.36	0.45	0.47	0.71	0.59	0.44	0.34	0.30	0.20	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page